



College of Basic and Applied Sciences
School of Engineering Sciences · Department of Computer Engineering

CPEN 208 — Project 1

Relational Database & Web Application
for the Computer Engineering Department

Name: David Kwame Odoi-Anim

Student ID: 22312110

Course: CPEN 208 — Introduction to Software Engineering

Semester: Second Semester, 2025/2026

Lecturer: Mr. Asiamah

PostgreSQL

Next.js 14

TypeScript

Tailwind CSS

GitHub: <https://github.com/doxaodoi/project-1-2>

1. Overview

This project delivers a relational database for a software the Computer Engineering Department intends to build, plus a Next.js 14 web application. The five required items are implemented as **functionalities users perform**, split across a **student** role and a departmental **admin** role:

#	Functionality	Who	Action
1	Student personal information	Student	View record and edit contact details
2	Student fees payments	Student	View statement and make a payment
3	Course enrollment	Student	Register / drop current-semester courses
4	Lecturers to course assignment	Admin	Assign / remove a lecturer on a course
5	Lecturers to TA assignment	Admin	Assign / remove a TA under a lecturer

The admin also views the whole **class roster with each student's outstanding fees**, which is exactly what the required database function `finance.get_outstanding_fees()` produces. Students additionally see their grades and a read-only view of who teaches their courses.

1.1 Assumptions

- Database `coe_dept`, four schemas: `people`, `academic`, `finance`, `auth`.
- Sample data is the real CPEN 208 class roster (70 students); emails follow `<studentID>@st.ug.edu.gh`.
- Two terms are seeded: **Semester 1 2025/2026** completed (graded) and **Semester 2 2025/2026** current (registerable).
- Each student is billed three items totalling **GH¢12,300.00**; payments vary.
- Accounts: every student uses password `Student@123`; a department admin signs in with `admin@coe.ug.edu.gh` / `Admin@123`.

2. Database Design

Four schemas hold the tables; every relationship is enforced with a foreign key. The `auth.users` account links to a student (or is an admin, with a null student link).

people

programs

program_id (PK), code, name, degree

students

student_id (PK), full_name, email, phone,
date_of_birth, level, program_id → programs

lecturers

lecturer_id (PK), staff_no, title, full_name, email,
academic_rank

teaching_assistants

ta_id (PK), full_name, email, student_id → students

academic

courses

course_id (PK), code, title, credit_hours, level,
semester

enrollments

student_id → students, course_id → courses,
academic_year, semester, grade

lecturer_course_assignment

lecturer_id → lecturers, course_id → courses,
academic_year, semester

lecturer_ta_assignment

lecturer_id → lecturers, ta_id → teaching_assistants,
course_id → courses

finance

fee_bills

student_id → students, academic_year, semester,
description, amount_due

payments

student_id → students, amount, paid_on, method,
reference

auth

users

user_id (PK), student_id → students (nullable, for
admins), full_name, email (unique), password_hash,
role

2.1 Build scripts (run in pgAdmin, in order)

`00_create_database` → `01_schemas` → `02_tables` → `03_functions` → `04_seed` (70 students, both terms, admin account) → `05_verify`. A database backup is included for one-step restore.

3. The Outstanding-Fees Function

The required function returns every student's `billed - paid` as a JSON array; the admin's Class & Fees screen is built on it.

```
CREATE OR REPLACE FUNCTION finance.get_outstanding_fees()
RETURNS JSON LANGUAGE sql STABLE AS $$
  SELECT COALESCE(json_agg(row_to_json(t) ORDER BY t.full_name), '[]'::json)
  FROM (
    SELECT s.student_id, s.full_name, s.email,
           COALESCE(b.total_billed,0) AS total_billed,
           COALESCE(p.total_paid,0) AS total_paid,
           COALESCE(b.total_billed,0) - COALESCE(p.total_paid,0) AS outstanding
    FROM people.students s
    LEFT JOIN (SELECT student_id, SUM(amount_due) total_billed
               FROM finance.fee_bills GROUP BY student_id) b ON
    b.student_id=s.student_id
    LEFT JOIN (SELECT student_id, SUM(amount) total_paid
               FROM finance.payments GROUP BY student_id) p ON
    p.student_id=s.student_id
    ) t;
$$;
```

```
[ { "student_id":22312110, "full_name":"David Kwame Odoi-Anim",
    "total_billed":12300.00, "total_paid":9000.00, "outstanding":3300.00 } /* ...70
total */ ]
```

4. The Student Application

Next.js 14 (App Router, TypeScript, Tailwind). Login is role-aware — students land on their dashboard, admins on the admin console. Authentication uses a JWT in an HttpOnly cookie.

CoE Department Portal
Computer Engineering · University of Ghana

Sign in

Email

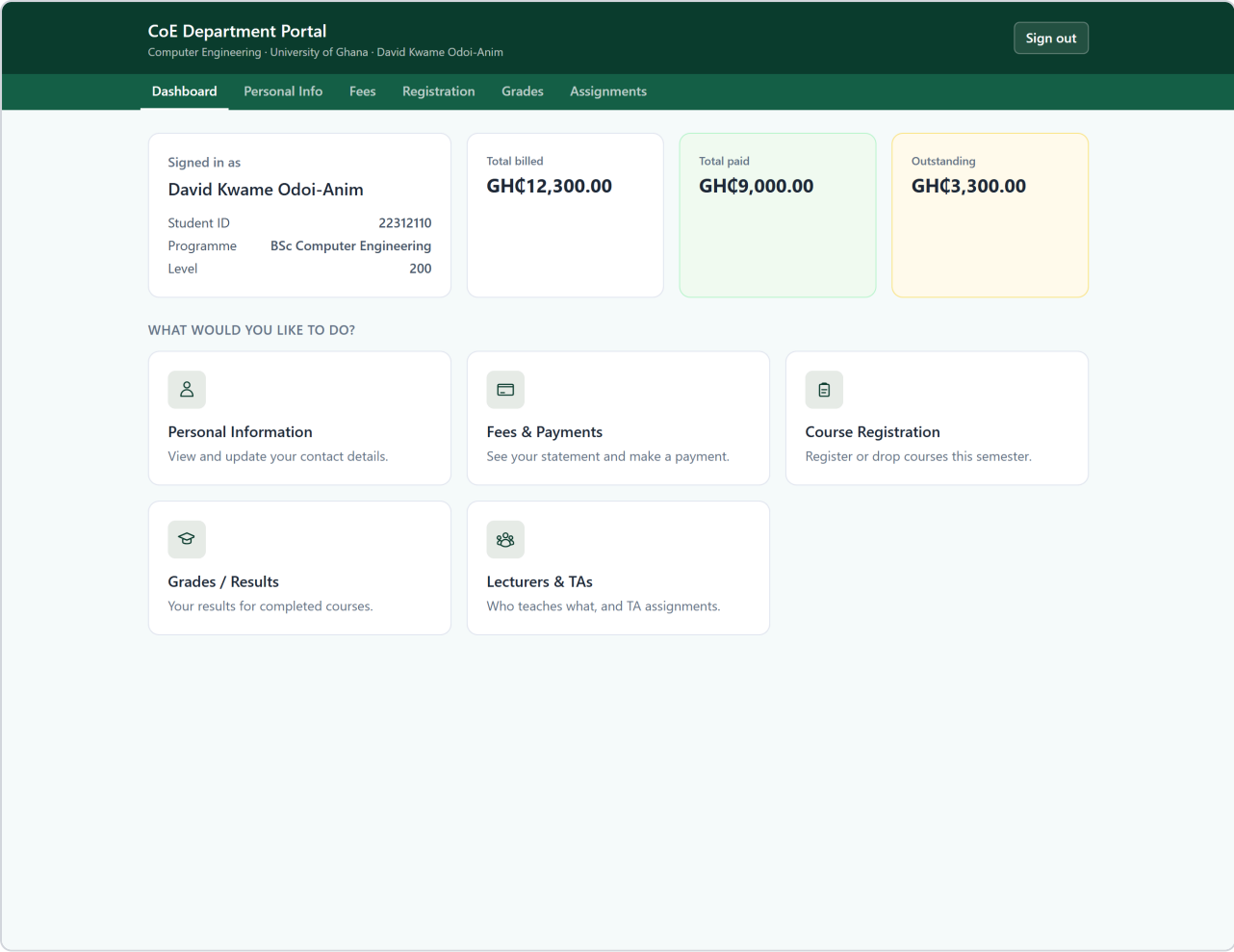
Password

Sign in

No account? **Register**

Demo: any seeded student e.g. 22312110@st.ug.edu.gh / Student@123

Login — role-aware; routes students and admins to the right area.



Student dashboard — fee summary and a launch-pad to each functionality.

4.1 Personal information & Fees (functionalities 1 & 2)

CoE Department Portal
Computer Engineering - University of Ghana - David Kwame Odoi-Anim

Sign out

DashboardPersonal InfoFeesRegistrationGradesAssignments

Personal Information

READ-ONLY DETAILS

Full name	David Kwame Odoi-Anim	Student ID	22312110
Programme	BSc Computer Engineering	Level	200
Date of birth	2004-04-26		

EDITABLE CONTACT DETAILS

Email

22312110@st.ug.edu.gh

Phone

+233241703923

Save changes

Personal information — identity details plus an editable contact form.

CoE Department Portal

Computer Engineering - University of Ghana - David Kwame Odoi-Anim

Sign out

Dashboard

Personal Info

Fees

Registration

Grades

Assignments

Fees & Payments

Total billed

GH¢12,300.00

Total paid

GH¢9,000.00

Outstanding

GH¢3,300.00

MAKE A PAYMENT

Amount (GHS)

up to 3300.00

Method

Bank

▼

Pay now

Payment History

Date	Amount	Method	Reference
2026-06-12	GH¢6,000.00	BANK	PAY-22312110-1
2026-07-14	GH¢3,000.00	MOMO	PAY-22312110-2

Fees — statement, a payment form (balance drops on success), and history.

4.2 Course registration & grades (functionality 3)

CoE Department Portal

Computer Engineering - University of Ghana - David Kwame Odoi-Anim

Sign out

Dashboard

Personal Info

Fees

Registration

Grades

Assignments

Course Registration

Second Semester, 2025/2026

Registered courses

2 course(s) - 6 credits

Code	Title	Credits	Lecturer	Action
CPEN 208	Introduction to Software Engineering	3	Prof. Elsie Adjei	Drop
CPEN 212	Digital Systems Design	3	Dr. Efua Danso	Drop

Available courses

Code	Title	Credits	Lecturer	Action
CBAS 210	Academic Writing II	3	Dr. Ama Boateng	Register
CPEN 214	Signals and Systems	3	Dr. Yaw Owusu	Register
CPEN 216	Electronic Circuits	3	Dr. Kwame Mensah	Register
MATH 224	Linear Algebra	3	Mr. Kofi Antwi	Register

Registration — register/drop current-semester courses (writes to enrollments).

Grades / Results

Completed courses

Indicative GPA

3.08

Code	Title	Credits	Year	Sem	Grade
CBAS 208	Academic Writing I	3	2025/2026	1	C
CPEN 202	Object Oriented Programming	3	2025/2026	1	B
CPEN 204	Data Structures and Algorithms	3	2025/2026	1	A
CPEN 206	Linear Circuits	3	2025/2026	1	A
CPEN 210	Digital Logic Design	3	2025/2026	1	B
MATH 221	Calculus II	3	2025/2026	1	C+

GPA is indicative, computed from published grades on a 4.0 scale.

Grades — results for completed courses, with an indicative GPA. Students also have a read-only "Assignments" page showing who teaches each course.

5. The Admin Console

The department admin manages the assignment functionalities and oversees class fees.

Class & Fees

70 students · balances from the `get_outstanding_fees()` database function.

Student ID	Name	Email	Billed	Paid	Outstanding
22384451	Abu Neaquittae Golda	22384451@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22376708	Adjei Priscilla	22376708@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22412982	Adu Mini	22412982@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22357814	Adzasa Stephen Yaw	22357814@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
22375367	Afia Beaa Osei-Safo	22375367@st.ug.edu.gh	GH¢12,300.00	GH¢5,000.00	GH¢7,300.00
22397756	Agbemavi Ryan	22397756@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22369321	Agormeda Nathaniel Tetteh	22369321@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22301848	Ahmad Mohammed Sahih Kayelgu	22301848@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22339520	Amprofi Yaa Obeng	22339520@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
22333597	Asante Esme Lilian	22333597@st.ug.edu.gh	GH¢12,300.00	GH¢5,000.00	GH¢7,300.00
22268986	Asante Gabriel Kwaku	22268986@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22381577	Botchway Daniel	22381577@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22315830	Brian Assibey-Yeboah	22315830@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22388189	Caleb Mensah	22388189@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
22393520	Cyril Desmond Ofori	22393520@st.ug.edu.gh	GH¢12,300.00	GH¢5,000.00	GH¢7,300.00
22407018	Daniel Dwomoh Frimpong	22407018@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22312110	David Kwame Odoi-Anim	22312110@st.ug.edu.gh	GH¢12,300.00	GH¢9,000.00	GH¢3,300.00
22271867	De-Andra Rebecca Ayebo	22271867@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
22300896	Doe Collins Kweku	22300896@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22402666	Dogbatse Darlington	22402666@st.ug.edu.gh	GH¢12,300.00	GH¢5,000.00	GH¢7,300.00
22397491	Douglas Kwaw Adjei	22397491@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22387715	Dzidzor Apu Apawudza	22387715@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
22382302	Edward Kakra Ankrah	22382302@st.ug.edu.gh	GH¢12,300.00	GH¢5,000.00	GH¢7,300.00
22382601	Eklou Juliet	22382601@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22379061	Emmanuel Akotuah Osae	22379061@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22368809	Emmanuel Dery	22368809@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22370498	Ethan Edric Kweku Nartey	22370498@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22328334	Etsey Hannah Seyram	22328334@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22321110	Gideon Nana Osei Amofa	22321110@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
22382425	Gilbert Akwasi Sarkodie Yeboah	22382425@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
22396551	Jerrold Xornam Kyekye	22396551@st.ug.edu.gh	GH¢12,300.00	GH¢5,000.00	GH¢7,300.00
22398562	Joseph Amankwah	22398562@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22398596	Joshua Appiah	22398596@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22394866	Joshua Kwame Asirifi	22394866@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22385323	Jude Gyampoh Addo	22385323@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22415339	Kelvin Kwesi Saah	22415339@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22303421	Kemausuor Winambe Tetteh-Kumah	22303421@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
22407033	Kenzi Segbefia	22407033@st.ug.edu.gh	GH¢12,300.00	GH¢5,000.00	GH¢7,300.00
22299189	Kessey Ntiako David	22299189@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22407837	Kingsley Caldicock Quartey	22407837@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22412615	Kofi Boateng Oware-Tano	22412615@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22411009	Kwaku Aninkorah Barimah	22411009@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
22382547	Kwame Ayeh Obeng	22382547@st.ug.edu.gh	GH¢12,300.00	GH¢5,000.00	GH¢7,300.00
22373317	Kwamena Kesse Quaicoe	22373317@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22395074	Lydia Tiwaah	22395074@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22339058	Maame Abena Amihere Ahu	22339058@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22302628	Maame Araba Grant-Aidoo	22302628@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00

22396566	Manford Kelvin Oppong	22396566@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
224018189	Mas'ud Nasir	224018189@st.ug.edu.gh	GH¢12,300.00	GH¢5,000.00	GH¢7,300.00
22385391	Najiib Abdul-Majeed Stephen	22385391@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22325819	Nana Adwoa Dansowaah Odoom	22325819@st.ug.edu.gh	GH¢12,300.00	GH¢5,000.00	GH¢7,300.00
22344703	Nana Anokye	22344703@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22306910	Newlove Yeboaah Kwarfo	22306910@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22385472	Obeng Ernest Antwi	22385472@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22399214	Obeng Ruth	22399214@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
22400543	Ocansey Frederick	22400543@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
22263126	Owusu Koranteng Yaw Poku	22263126@st.ug.edu.gh	GH¢12,300.00	GH¢5,000.00	GH¢7,300.00
22373463	Owusu Nana Boadiwaa	22373463@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22306021	Paul Badu Amponsah	22306021@st.ug.edu.gh	GH¢12,300.00	GH¢5,000.00	GH¢7,300.00
22381702	Paula Akosua Asiedua Frimpong	22381702@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22387846	Quaicoo Emile	22387846@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22377537	Reuben Adomako	22377537@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22263922	Romel Alvin Nii Lartey Lartey	22263922@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
22401641	Sandra Naa Adaku Mettle	22401641@st.ug.edu.gh	GH¢12,300.00	GH¢5,000.00	GH¢7,300.00
22403781	Sekyere Kofi Bempong	22403781@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22304260	Tetteh Christian Edward Nii Mantey	22304260@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22304013	Tietaah Sonnu	22304013@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22416112	Troy Thomas	22416112@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22302188	Van Jerry Quansah	22302188@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
22299949	William Enchill	22299949@st.ug.edu.gh	GH¢12,300.00	GH¢5,000.00	GH¢7,300.00

Class & Fees — the full roster with each student's billed / paid / outstanding, powered by `finance.get_outstanding_fees()`.

Lecturer & TA Assignments

Lecturer → Course (functionality 4)

Lecturer	Course	Semester	
Select lecturer	Select course	2	Assign

Course	Title	Lecturer	Sem	
CBAS 208	Academic Writing I	Prof. Elsie Adjei	1	Remove
CPEN 202	Object Oriented Programming	Dr. Kwame Mensah	1	Remove
CPEN 204	Data Structures and Algorithms	Dr. Ama Boateng	1	Remove
CPEN 206	Linear Circuits	Dr. Yaw Owusu	1	Remove
CPEN 210	Digital Logic Design	Dr. Efua Danso	1	Remove
MATH 221	Calculus II	Mr. Kofi Antwi	1	Remove
CBAS 210	Academic Writing II	Dr. Ama Boateng	2	Remove
CPEN 208	Introduction to Software Engineering	Prof. Elsie Adjei	2	Remove
CPEN 212	Digital Systems Design	Dr. Efua Danso	2	Remove
CPEN 214	Signals and Systems	Dr. Yaw Owusu	2	Remove
CPEN 216	Electronic Circuits	Dr. Kwame Mensah	2	Remove
MATH 224	Linear Algebra	Mr. Kofi Antwi	2	Remove

Lecturer → Teaching Assistant (functionality 5)

Lecturer	Teaching Assistant	Course	
Select lecturer	Select TA	Select course	Assign

Lecturer	TA	Course	
Dr. Ama Boateng	Selorm Agbeko	CPEN 204	Remove
Dr. Efua Danso	Linda Ofori	CPEN 212	Remove
Prof. Elsie Adjei	Akosua Frimpong	CPEN 208	Remove
Prof. Elsie Adjei	Nana Kwabena Asare	CPEN 208	Remove

Assignments — assign/remove **lecturer**→**course** (functionality 4) and **lecturer**→**TA** (functionality 5). Each action writes to the database.

Course Catalogue

ADD A COURSE

Code	Title		
CPEN 220	Course title		
Credits	Level	Semester	Add course
3	200	2	

Code	Title	Credits	Level	Sem
CBAS 208	Academic Writing I	3	200	1
CPEN 202	Object Oriented Programming	3	200	1
CPEN 204	Data Structures and Algorithms	3	200	1
CPEN 206	Linear Circuits	3	200	1
CPEN 210	Digital Logic Design	3	200	1
MATH 221	Calculus II	3	200	1
CBAS 210	Academic Writing II	3	200	2
CPEN 208	Introduction to Software Engineering	3	200	2
CPEN 212	Digital Systems Design	3	200	2
CPEN 214	Signals and Systems	3	200	2
CPEN 216	Electronic Circuits	3	200	2
MATH 224	Linear Algebra	3	200	2

Courses — the catalogue with a form to add new courses.

6. Non-Functional Requirements

Beyond the five functionalities, the software meets the quality requirements expected of a department system:

Category	How it is addressed
Security	BCrypt password hashing; stateless JWT (HS256); HttpOnly cookie; role-based access (STUDENT/ADMIN) plus per-record ownership checks; fully parameterized SQL (no injection); the database password is supplied via an environment variable and never committed.
Validation & error handling	Server-side validation (positive payments not exceeding the balance, required fields, valid semester); consistent JSON error responses with correct status codes (400/401/403/404/409).
Performance	Indexes on foreign-key/lookup columns; HikariCP connection pooling; a stateless API that scales horizontally; no-cache reads keep data fresh.
Reliability & integrity	Foreign-key and UNIQUE constraints; a transactional registration path; re-runnable, idempotent SQL scripts; a database backup is provided.
Usability & accessibility	Responsive layout; labelled form controls; clear success/error feedback; role-appropriate navigation.
Maintainability	Layered controller → service → repository design; DTO records; centralized configuration; small, single-purpose modules.
Portability	Standard PostgreSQL; environment-driven config with an <code>.env.example</code> ; runs locally without containers.

7. How to Run

1. **Database:** in pgAdmin run `00` on `postgres`, then `01 – 05` on `coe_dept` (or restore `coe_dept_backup.backup`).
2. **API (Project 2):** `cd "PROJECT 2/api"`, set `DB_PASSWORD`, `mvn spring-boot:run` (port 8080).
3. **Front end:** `cd "PROJECT 1/web"`, copy `.env.example` → `.env.local`, `npm install && npm run dev`, open `http://localhost:3000`.
4. Student: `22312110@st.ug.edu.gh` / `Student@123`. Admin: `admin@coe.ug.edu.gh` / `Admin@123`.

Deliverables: SQL scripts + backup in `PROJECT 1/database`; front-end source in `PROJECT 1/web`; full source on GitHub at <https://github.com/doxaodoi/project-1-2>.